

Sulfite sensitivity: significance in human health.

Endogenous sulfite is generated as a consequence of the body's normal processing of sulfur-containing amino acids. Sulfites occur as a consequence of fermentation and also occur naturally in a number of foods and beverages. As food additives, sulfiting agents were first used in 1664 and approved in the United States as long ago as the 1800s. With such long experience with their use, it is easy to understand why these substances have been regarded as safe. They are currently used for a variety of preservative properties, including controlling microbial growth, preventing browning and spoilage, and bleaching some foods. It is estimated that up to 500,000 (< .05% of the population) sulfite-sensitive individuals live in the United States. Sulfite sensitivity occurs most often in asthmatic adults--predominantly women; it is uncommonly reported in preschool children. Adverse reactions to sulfites in nonasthmatics are extremely rare. Asthmatics who are steroid-dependent or who have a higher degree of airway hyperreactivity may be at greater risk of experiencing a reaction to sulfite-containing foods. Even within this limited population, sulfite sensitivity reactions vary widely, ranging from no reaction to severe. The majority of reactions are mild. These manifestations may include dermatologic, respiratory, or gastrointestinal signs and symptoms. Severe nonspecific signs and symptoms occur less commonly. Broncho-constriction is the most common sensitivity response in asthmatics. The precise mechanisms of the sensitivity responses have not been completely elucidated. Inhalation of sulfur dioxide (SO₂) generated in the stomach following ingestion of sulfite-containing foods or beverages, a deficiency in a mitochondrial enzyme, and an IgE-mediated immune response have all been implicated.(ABSTRACT TRUNCATED AT 250 WORDS)